APPROVED
By Heather Campbell at 2:49 pm, 2/25/25

TUBE SUPPLY

4669 Brittmoore Road
Houston, TX 77041
Tel. 713-466-4130
Fax 713-856-9114

Certificate of Conformance

Order Date	2/25/2025
Shipping Order	01146272
Customer	CACTUS WELLHEAD LLC
Purchase Order	TRP0049110
Part Number	119134 REV A03
Specification	CMS-107 REV H, CMS-002 REV L, CMS-004 REV C, CIR-002 REV F
Grade	HR 4140 110 KSI
Size	7.250 OD Round Bar
Heat Number	4000025514
Lot Number	400002551484
Quantity	1 @ 18'-10" E/L

1) Triple-S Tube Supply LP, dba Tube Supply certifies that the above referenced material complies with the terms of the order contract, as presently agreed upon.

2) This order was processed in accordance with the Triple-S Tube Supply ULC, dba Tube Supply, Quality Manual, QM-01 Revision H, dated 6/5/23.

3) No weld repair was performed on this material

Other Comments:

Patty Garza
Account Manager
Tube Supply

NUCOR®

Mill Certification

12/16/2024

MTR#:1841463-8
Lot #:400002551484
3601 Paul R Lowry Road
Memphis, TN 38109 US
901 786-5900
Fax: 901-786-5901

Sold To: TREK METALS INC
5221 N OCONNOR BLVD
STE 750
IRVING, TX 75039 US

Ship To: TREK METALS- RENTER MAT. HAND
12426 TICONDEROGA RD
HOUSTON, TX 77044 US

APPROVED
SO
16-Jan-25

Customer PO	3534	Sales Order #	40038897 - 1.2
Product Group	Hot Roll - Engineered Bar	Product #	1107020
Grade	4140/42 110 KSI	Lot #	400002551484
Size	7.25"	Heat #	4000025514
BOL #	BOL-1887390	Load #	1841463
Description	Hot Roll - Engineered Bar Round 7.25" (7 1/4") 4140/42 110 KSI 20' Random Normalize Heat Treat Quench and Tempered Ultrasonically Tested Machine Surface Inspected 6001-10000 lbs	Customer Part #	
Production Date	12/09/2024	Qty Shipped LBS	37670
Product Country Of Origin	United States	Qty Shipped EA	14
Original Item Description		Original Item Number	

I hereby certify that the material described herein has been manufactured in accordance with the specifications and standards listed above and that it satisfies those requirements.

Melt Country of Origin : United States — Melting Date: 08/17/2024

Roll Country of Origin : United States — Rolling Date: 10/16/2024

C (%)	Mn (%)	P (%)	S (%)	Si (%)	Cu (%)	Ni (%)	Cr (%)	Mo (%)	V (%)	Al (%)	B (%)
0.41	0.94	0.009	0.012	0.272	0.19	0.22	1.07	0.23	0.025	0.028	0.0004

Sn (%)	Ti (%)	Nb (%)	Ca (%)	Pb (%)	As (%)	Bi (%)	Sb (%)	Te (%)	N (PPM)	H (PPM)
0.006	0.002	0.003	0.0010	0.000	0.004	0.001	0.001	0.000	57	1

DI Calculated (IN) : 7.16

Sulfur + Phosphorus (%) : 0.021

Austenitic fine grain, size 5 or finer, by chemical analysis per the latest revision of ASTM A29.
Reduction Ratio 7.70 : 1

ASTM E45 Method A (Worst)

(1)	Sulfides T: 1.5	H: 1.5	Alumina T: 1.0	H: 0.0	Silicates T: 0.0	H: 0.0	Globular T: 1.0	H: 0.5

E381 Macroetch

	Macroetch E381 Surface	Macroetch E381 Mid Radius	Macroetch E381 Center
(1)	1	1	1

Heat Treatment

	Normalizing Temp (FAH)	Normalizing Time (Min)	Austenitizing Temp (FAH)	Austenitizing Time (Min)	Tempering Temperature (FAH)	Tempering Time (Min)
(1)	1617	69	1605	56	1108	75

Hardness

	Brinell (HBW)
(1) Surface	308
(1) 1" Below	305

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Hardness

	Brinell (HBW)
(1) Mid-radius	314

Tensile testing

	Yield 0.2% Offset (PSI)	Tensile (PSI)	Elongation in 2" (%)	ROA (%)	Yield 0.6% EUL (PSI)
(1) 1" Below	134000	150000	17	56	133000
(1) Mid-radius	131000	151000	16	46	131000

Charpy Test Results

Temperature	Orientation	Location	Specimen	Impact Energy, ft•lbf	Lat EXP, In.	%Shear
70F	Longitudinal	1" Below	1	54	0.030	100
70F	Longitudinal	1" Below	2	55	0.029	100
70F	Longitudinal	1" Below	3	54	0.030	100
			Average	54	0.030	100

Temperature	Orientation	Location	Specimen	Impact Energy, ft•lbf	Lat EXP, In.	%Shear
14F	Longitudinal	1" Below	1	49	0.032	100
14F	Longitudinal	1" Below	2	50	0.023	100
14F	Longitudinal	1" Below	3	50	0.020	100
			Average	50	0.025	100

Temperature	Orientation	Location	Specimen	Impact Energy, ft•lbf	Lat EXP, In.	%Shear
-20F	Longitudinal	1" Below	1	34	0.019	10
-20F	Longitudinal	1" Below	2	33	0.019	10
-20F	Longitudinal	1" Below	3	34	0.022	10
			Average	34	0.020	10

Comments:
4140 HOT ROLLED/QT BAR 110 KSI MINIMUM YIELD
COMPLIANT WITH ASTM A29, A322, A434 CL BD LATEST REVISIONS

FURNACE TYPE: CONTINUOUS / TEMPERATURE MONITORED BY INFRARED PYROMETER.
MATERIAL IS STRESS FREE.
MECH TESTS TAKEN FROM PROLONGATION
POST HEAT TREAT, BAR STOCK 100% UT INSPECTED TO API 7-1,SEC. 11/ASTM A388/API 6A PSL 3, RESULT=PASSED
HEAT TREAT FURNACES CALIBRATED PER MANUFACTURER'S SPECIFICATIONS
QUENCH MEDIA: WATER (UNLESS OTHERWISE STATED IN THE COMMENTS). IF QUENCH MEDIA IS WATER, THE QUENCH MEDIA START TEMPERATURE DID NOT EXCEED 100F AND THE FINISH TEMPERATURE DID NOT EXCEED 120F.

1. Melt/cast practice: electric arc furnace melted, ladle refined, vacuum degassed, continuously cast.
2. Nucor Steel Memphis is compliant with the Pressure Equipment Directive (PED) 2014/68/EU 7/2, Annex I, Paragraph 4.3.
3. All manufacturing processes, including melting, have been performed in the U.S.A.
4. No welding or weld repair was performed on this product.
5. No mercury, mercury compounds or mercury containing devices came into contact with this product.
6. Product verified not to exceed background radiation levels.
7. In regards to Section 1502 of the Dodd-Frank Act ("Conflict Minerals"), no covered minerals (tantalum, tin, tungsten or gold) were intentionally added to this product.
8. This product conforms to the specifications described on this document. This document may not be reproduced except in full, without written approval of Nucor Corporation.
9. If applicable to the product certified by this report, test procedures are performed in compliance with the current revisions (at the time of testing) of the following ASTM standards:
chemical analysis: E415/E1019, grain size: E112, macrostructure/macroetch: E381, tensile testing: E8/A370, hardness testing: E10 (Type A) /E18/A370. Charpy impact: E23/A370, microcleanliness: E45, decarburization depth: E1077, Jominy: A255.
10. Reported results for ASTM E45, E1077, and E112 are provided as an interpretation of ASTM procedures.
11. If applicable to this report, ASTM E23 tests conducted with 8mm radius striker and 10mm x 10mm V-notch specimen, unless otherwise stated.
12. This document complies with EN 10204 "type 3.1".
13. The following tests are outside the ISO 17025 Laboratory Scope for Nucor Steel Memphis: hydrogen testing and non-destructive testing.
14. Tests not specifically listed in notes 9 and 13, if applicable to this report, are performed by ISO 17025 certified providers outside of Nucor Steel

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Memphis. Unless otherwise noted, any tests performed by certified providers are within the specific provider's ISO 17025 scope.



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